

Manual

Measurement Data Interface for Quality Data AIP-MDI 8.2

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Contents

1	Overview – Measurement Data Interface for Quality Data	4
2	Overview of Measurement Data Interface for Quality Data	5
3	MDI Configuration	6

1 Overview – Measurement Data Interface for Quality Data

Purpose

This component permits the automated collection of inspection data in conjunction with a user-initiated action.

Implementation Notes

The component is used when interface devices are required to be connected to interfaced inspection equipment for automated collection of inspection data.

Integration

This component serves components "Recording/information functions for quality data" and extends the data collection functions of the AIP shop floor client.

Features

The following functions are available:

- Automatic or triggered (e.g. by pressing a pedal) collection of measured values from inspection systems (caliper gage, micrometer, measuring mandrel, measuring microscope, etc.) connected via an interface device or data interfaces.
- Characteristic-related automatic collection of measured values
- Measured value buffer to hold the measured values collected by the measuring equipment drivers

The function within the context of collection with the AIP is described in the [AIP-CAQ manual](#).

2 Overview of Measurement Data Interface for Quality Data

Application	Document	Transaction code	Function authorization
MDI Configuration	MOC_MeasurementDataInterface.pdf	mdi	mdi

3 MDI Configuration

Summary

Menu	Master data → Quality management → MDI configuration
Transaction code	mdi
Function authorization	mdi

MDI stands for **M**easurement **D**ata **I**nterface.

The "MDI configuration" application is the basis for online data collection. However, the configuration varies subject to the measurement equipment/interface devices/measuring systems to be connected. The configurations made here allow for the measured values recorded by the individual MDI drivers to be transferred to the individual characteristics of the inspection orders.

Different MDI drivers are used, depending on the measurement equipment/interface device/measuring system to be connected. The below list indicates some of the available MDI drivers.

- MDI serial (for connections via serial interfaces. Only one measured value is transferred with each serial transfer)
- MDI measured value list (a file includes measured values of several characteristics. Each characteristic is written in a separate line/row)
- MDI measured value file (a file includes measured values of several characteristics. Each characteristic is written in a separate column)
- MDI serial list (for connections via serial interfaces. Each serial transfer can send measured values for several characteristics at the same time. The different characteristics are indicated in separate lines)
- MDI Steinwald (special driver to connect interface devices by STEINWALD datentechnik GmbH)
- MDI Ibrit (special driver to connect interface devices by IBR – Messtechnik GmbH)

Utilization

The "Measurement equipment connection" field includes the ID number of the test equipment configuration to be created. Any short description of the measurement equipment connection may be entered in the "Description" field. Ideally, the channel description matches the description of the MDI driver. The "channel" field includes the channel number of the MDI driver used for recording measured values. Once the configuration has been completed, it has to be released by checking the "active" field.

In the "connection" tab you need to enter the IP address of the computer on which the MDI driver was started. Use the IP address 127.0.1, in the event that the MDI driver is running on the same computer that is also supposed to receive the measured values from the connected measurement equipment (i.e. where measurement value recording will take place).

The TCP port number of the computer on which the MDI driver was started has to be entered in the "communication" tab. Another port value than the default port value 9900 is usually only required if several MDI drivers are running on one computer.

Specific settings are made in the "properties" tab. The functions of these fields are explained in the below field description.

Integration

This function has been integrated in the measurement recording function and allows for measured values to be received/collected automatically. In addition, this application allows for resources of the "PRM" type (gages) to be assigned to an MDI configuration. A basis for online data collection is established, as resources of the "PRM" type are assigned to the characteristics/inspection plan characteristics.

Prerequisite

To be able to use this function in a reasonable manner, resources of the "PRM" type have to be created beforehand, to be in the position to assign them to the corresponding MDI configurations.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

Measurement equipment connection

ID number of the MDI configuration

Designation

Description of the MDI configuration

Channel

Channel number that has been defined in the corresponding MDI driver.

active

The MDI configuration is released and available, once the "active" field has been checked.

IP address

IP address of the computer on which the MDI driver is installed. The IP address 127.0.0.1 is to be entered if this is the local computer.

Port

Communication port, by default 9900. If this port is already assigned the next higher port number is to be entered.

Performance/behavior during measurement recording

- Measured values are recorded, irrespective of the characteristic that is active.

Within measurement recording measured values of all characteristics can be read out of the MDI drivers and assigned to the inspection order characteristics at once. Consequently, this configuration does not require the corresponding characteristic to be active in measurement recording to be able to transfer the respective measured values. This configuration has to be selected, e.g. for complex measuring equipment delivering values for several characteristics in one measurement at the same time.

- Measured values are only collected if the corresponding characteristic is active.

In measurement recording only measured values for the currently selected characteristic are read out from the MDI driver and assigned to it. In the majority of cases, this configuration is used for locally connected measurement equipment, e.g. sliding calipers.

- Enter measured values directly into the input field

If this option is checked, the measured value recorded by the MDI driver is directly copied to the entry field of the selected characteristic.

Delete measured value buffer at activation

If this option is checked, the measured value buffer for this channel is deleted on the MDI server upon activation of the channel. This setting is mostly used with measurement devices for which the option "Measured values are only collected if corresponding characteristic is active" has been activated.

Delete measured values that cannot be saved (plausibility checking) from buffer

In case this option is checked, measured values that cannot be stored, e.g. due to not passing the plausibility/validity check, are deleted from the buffer for this MDI channel.

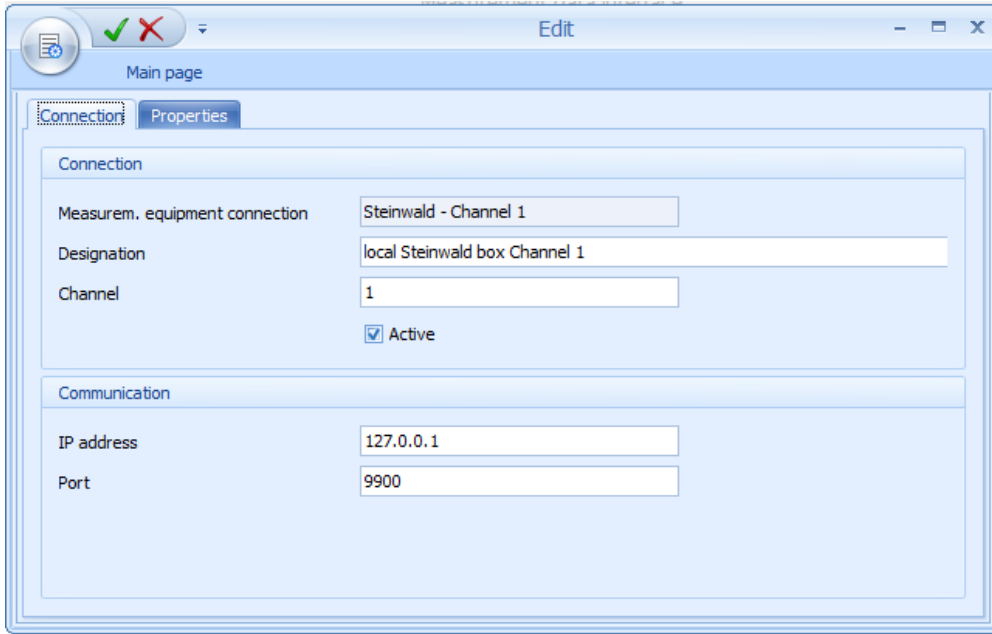
Additional parameters

This field may be used to enter additional configuration parameters to be transmitted to the specified MDI driver channel. Only in rare cases are additional configuration parameters required.

One example for configuring this parameter is specifying the interval for querying the corresponding test device channel. In order to be able to apply a configuration, the MDI driver and the measurement equipment connected to it must support the parameters defined here. Several parameters are separated by semicolon.

Editing functions

The following dialog opens to edit a data record:



The screenshot shows a software window titled "Edit". Inside, there's a "Main page" tab and a "Properties" sub-tab. The "Connection" section has the following fields:

- Measur. equipment connection: Steinwald - Channel 1
- Designation: local Steinwald box Channel 1
- Channel: 1
- Active: ☒

The "Communication" section has the following fields:

- IP address: 127.0.0.1
- Port: 9900

Toolbar



Entries

Assignment of resources

Function to assign resources of the type "PRM".